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# MATH SURVEY

## SPRING 1985

### VOL.59

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Noam Elkies and Kenny Yoon

# ISOPERIMETRIC THEOREM

The Isoperimetric Inequality states:

- Of all plane figures with a given perimeter, the circle has the greatest area.
- Of all plane figures with a given area, the circle has the least perimeter.

Another way of stating it is:  $4A < P^2$

There are many similar isoperimetric problems for polygons, three dimensions, etc.

Most people 'know' that the circle encloses the greatest area for a given perimeter; in fact, it was known to Archimedes in the second century B.C. However, it wasn't actually proven until the nineteenth century.

Jacob Steiner made a well-known, but unsuccessful attempt at proving the Isoperimetric Inequality. (Note: It doesn't seem as well-known that he didn't succeed.) He showed that for any plane figure which is not a circle there is another with greater area and the same perimeter. The following is essentially his argument:

If the figure is concave, we choose two points on the boundary of the figure such that the segment between them lies entirely outside the figure, and we reflect the portion of the figure between the two endpoints about the segment (see Fig. 1).

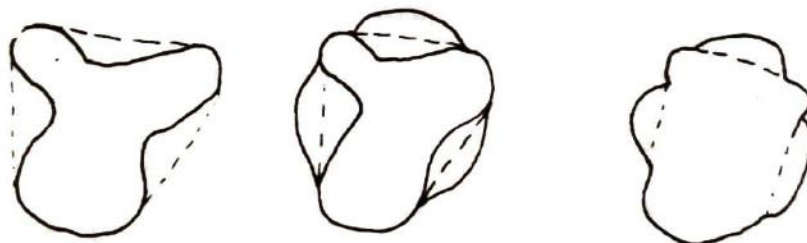


Fig. 1

If the figure is convex, we choose two points, A and B, on the boundary of the figure such that they divide the perimeter into two equal parts. If the area of the region on one side of the segment AB is greater than the area of the region on the other side, we make a new figure by reflecting the greater region about segment AB, as in Fig. 2.

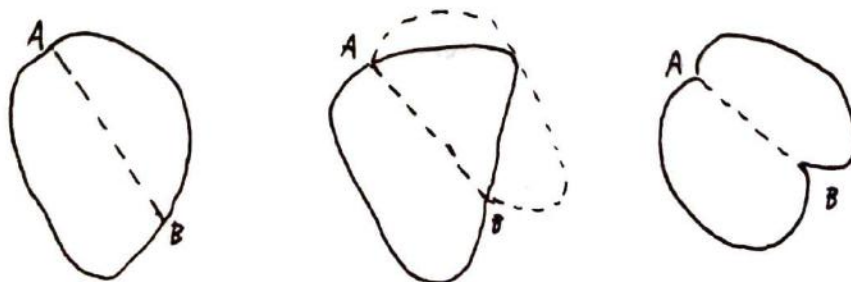


Fig. 2



Now consider the region on one side of segment  $AB$ . If it isn't a semicircle, we can find a point  $C$  on the boundary of the region such that  $\triangle ACB$  isn't a right triangle. We create a new half-figure with  $A_1=A_1$ ,  $A_2=A_2$ , and  $\triangle A'C'B'$  a right triangle. Since  $\triangle A'C'B'$  is a right triangle, and  $A'C'=AC$  and  $B'C'=BC$ , the area of  $\triangle ACB$  will be less than that of  $\triangle A'C'B'$ . Thus the area of the new half-figure will be greater, while its perimeter will remain the same. We reflect this about  $A'B'$  to get the complete new figure.

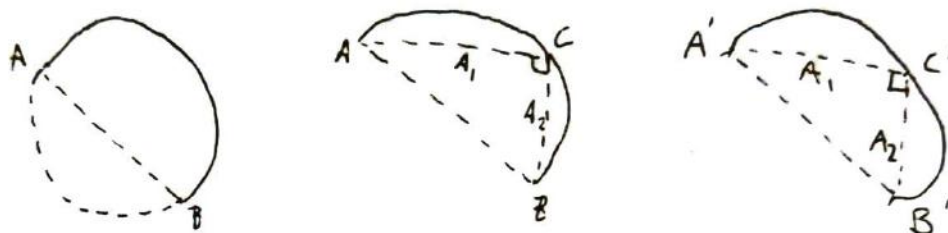


Fig. 3

Since it is possible to increase the area of any figure except for the circle without changing its perimeter using this method, the circle must have the greatest area for a given perimeter.

This completes Steiner's argument. Karl Weierstrass, who established the rigor of modern analysis, found the error in the proof by Steiner. The flaw is that he assumes that there must be a figure with a greatest area for a given perimeter. Intuitively, this seems true; it is obviously impossible for a figure with an area of one square foot say, to have a perimeter of one-sixteenth of an inch. However, it is not so simple to prove.

Weierstrass filled in the gap by supplying an indirect proof that such a figure must exist—that is, if such a figure did not exist there would be a contradiction. Almost all of Steiner's work had been done using only geometric, or constructive methods.

To solve this seemingly simple problem and other related ones, Weierstrass and others developed the branch of analysis known as the calculus of variations.

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# USEFUL TERMS

Professor Georg Friedrich Bernhard "Bud" Riemann, Math Survey's resident scholar, has graciously loaned his copy of this indispensable mathematical tool to 1985 Math Survey for publication. In accordance with his wishes, we would like to dedicate this glossary to his charges Little Jimmy, Little Janey, and their fellow neophyte mathematicians.

1. "It is easily shown that ..."  
(It'll take a ream of paper to prove it.)
2. "It can be shown that ..."  
(Let's skip the details.)
3. "It is clear that ..."  
(It can be proved in a finite number of steps.)
4. "It is obvious that ..."  
(You won't understand it, stupid.)
5. "Simplifying we find ..."  
(HAL can do it in three minutes flat.)
6. "To prove this, it is necessary to show ..."  
(This is the only way I know.)
7. "One may show ..."  
(One did. His name was Gauss.)
8. "It follows that ..."  
(It is not easy to show that ...)
9. "Clearly, ..."  
(The details are somewhat muddy.)
10. "It is well known that ..."  
(See "Mathematische Zeitschrift", vol. XXXVI, 1892,  
or "Xvixkundige Opgaven Met de Oplossingen, vol. 13,  
1919, Series B.)
11. "The student of trigonometry will notice that ..."  
(A math major might be able to prove ...)
12. "Finally, we have ..."  
(Only ten more steps to go.)
13. "Intuitively one can see that ..."  
(Only if your name is Nostradamus.)

(continued on page 17)



# GOLDBACH'S CONJECTURE

Goldbach's Conjecture asserts that every even number greater than two can be expressed as the sum of two primes; the stronger Goldbach's Conjecture says that every even number greater than seven can be expressed as the sum of two distinct primes. The proof of even the weaker form has yet to be accomplished, and is one of the famous unsolved problems of mathematics. We do not attempt to produce a proof here; rather, we present the following surprising result, whose proof can be said to provide probabilistic evidence for Goldbach's Conjecture.

Theorem: For every positive integer  $N$ , there exists a number which can be expressed as the sum of two primes in more than  $N$  ways.

proof: Consider  $f: [0,1) \rightarrow [0,\infty)$  defined by

$$f(x) = x^2 + x^3 + x^5 + \dots = \sum_{p \text{ prime}} x^p.$$

$[f(x)]^2 = \sum_{n=2}^{\infty} a(n)x^n$ , where  $a(n)$  is the number of ways of expressing  $n$  as the sum of two primes.

$$\text{Assume } [f(x)]^2 \leq \sum_{n=1}^{\infty} Nx^n = Nx/(1-x).$$

then  $f(x) \leq \sqrt{Nx/(1-x)}$ , and

$$\int_0^1 f(x) dx \leq \int_0^1 \sqrt{Nx/(1-x)} dx = \sqrt{N} \int_0^1 \sqrt{(1-y)/y} dy, \text{ where we have made the substitution } y=1-x.$$

But  $\int_0^1 \sqrt{(1-y)/y} dy < \int_0^1 \sqrt{1/y} dy = 2\sqrt{N}$ ; i.e., the improper integral converges.

However,  $\int_0^1 f(x) dx = \int_0^1 \sum_p x^p dx = \sum_p \int_0^1 x^p dx = \sum_p 1/(p+1) > \sum_p 1/2p = \frac{1}{2} \sum_p 1/p = \infty$ ; i.e., the improper integral diverges.

This contradiction establishes the theorem.

Q.E.D.

Math Survey is indebted for the statement and proof of this theorem to Michael Larsen, who presented it in a lecture at the Hampshire College Summer Studies in Mathematics, 1984.

# VECTOR GEOMETRY

We will show that  $i = \sqrt{-1}$  can be used to prove theorems in geometry. We assume that the reader is familiar with the fact that in the complex plane, multiplication of a vector by  $i$  is equivalent to a rotation of the vector 90 degrees in the counterclockwise direction. We will begin by proving a relatively simple theorem in geometry and then proceed to more difficult problems.

Theorem 1: The diagonals of a square are perpendicular

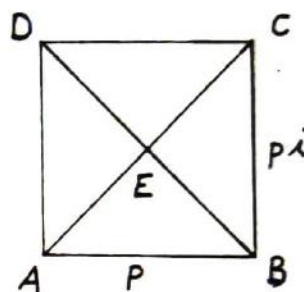
Proof: Let  $\vec{AB} = p$

$\vec{BC} = pi$ , as in Fig. 1

then  $\vec{AC} = p + pi$ , and  $\vec{BD} = pi - p$ . Fig. 1

$(\vec{AC})i = pi - p = \vec{BD}$

Q.E.D.



Theorem 2: If squares BAED and BCJK are constructed on sides AB and BC of  $\triangle ABC$ , then  $KA = CD$  and  $KA \perp CD$ .

Proof: Let  $\vec{BC} = p$

$\vec{CA} = q$

then  $\vec{KB} = pi$  and  $\vec{BA} = p + q$

$\vec{KA} = \vec{KB} + \vec{BA} = pi + (p + q)$

$\vec{AE} = \vec{BD} = (p + q)i$

$\vec{CD} = -p + (p + q)i$ , and

$(\vec{KA})i = -p + (p + q)i = \vec{CD}$

Q.E.D.

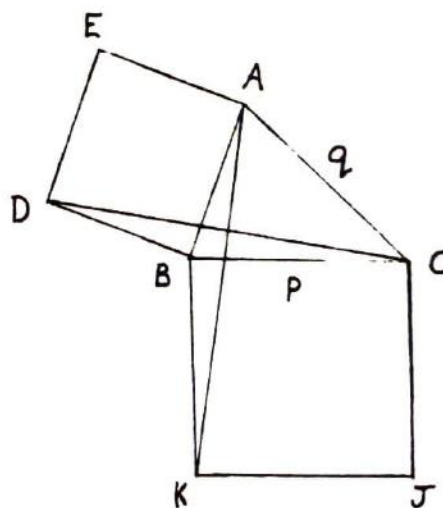


Fig. 2

We conclude with a theorem, which, to the best of our knowledge, is new.

Theorem 3: In triangle ABC, D is a point on AB, and E is a point on AC. If squares erected on BC and on DE have the same center, then  $\triangle ABC$  is a right triangle and  $BD = CE$ .



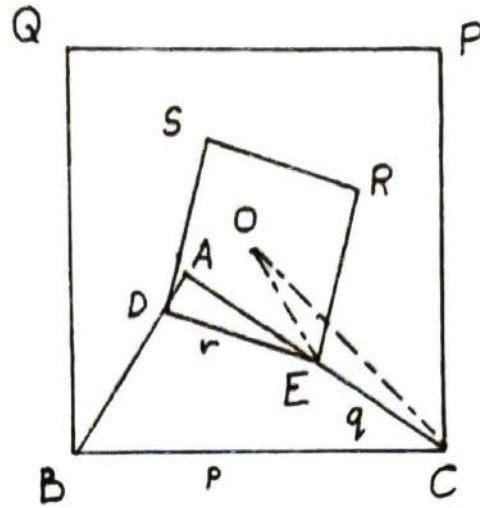


Fig. 3

Proof: Let  $\vec{BC} = p$

$$\vec{CE} = q$$

$$\vec{ED} = r$$

then  $\vec{BD} = p + q + r$ , as in Fig. 3.

Since O is the center of square DERS,

$$\vec{EO} = \frac{\sqrt{2}}{2}(\vec{DE})i^{3/2} = -\frac{\sqrt{2}}{2}ri^{3/2}.$$

Note that  $|\vec{EO}| = \frac{\sqrt{2}}{2}|\vec{DE}|$  and  $\vec{EO}$  is obtained by rotating  $\vec{DE}$  by  $135^\circ$  in the counterclockwise direction. Since  $i$  represents a rotation of  $90^\circ$ ,  $i^{3/2}$  represents a rotation of  $90^\circ + 45^\circ = 135^\circ$  (by DeMoivre's Theorem).

Similarly,  $\vec{CO} = \frac{\sqrt{2}}{2}(\vec{BC})i^{3/2} = \frac{\sqrt{2}}{2}pi^{3/2}$ , and

$$\vec{CE} + \vec{EO} = \vec{CO}.$$

$$\text{Thus, } q - \frac{\sqrt{2}}{2}ri^{3/2} = \frac{\sqrt{2}}{2}pi^{3/2}, \text{ or}$$

$$q = \frac{\sqrt{2}}{2}(p + r)i^{3/2} \quad (1)$$

In order to prove this theorem, we must show that

$$p + q + r = -qi.$$

Using the value of  $q$  obtained in (1), we have

$$(p + r)\frac{\sqrt{2}}{2}(p + r)i^{3/2} = -\frac{\sqrt{2}}{2}(p + r)i^{5/2}, \text{ or}$$

$$(p + r)\left[1 + \frac{\sqrt{2}}{2}i^{3/2} + \frac{\sqrt{2}}{2}i^{5/2}\right] = 0. \quad (2)$$

Now in polar form,

$$i = \cos 90^\circ + i\sin 90^\circ$$

$$i^{3/2} = \cos 135^\circ + i\sin 135^\circ$$

$$i^{5/2} = \cos 225^\circ + i\sin 225^\circ$$

Substituting the above in (2), we find that

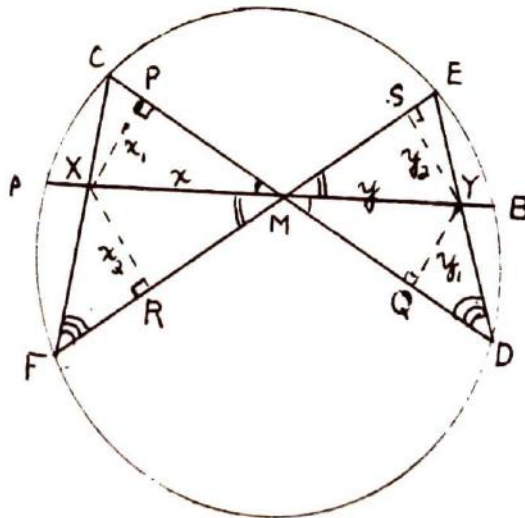
$$(p + r)[1 - \frac{1}{2} + \frac{1}{2}i - \frac{1}{2} - \frac{1}{2}i] = 0 \text{ identically.}$$

Q.E.D

# BUTTERFLY THEOREM

In solving mathematical problems, we are often faced with the question of whether or not two segments are equal. One of the multitude of theorems regarding the identification of congruent segments is the butterfly theorem.

**Theorem:** Let  $\overline{CD}$  and  $\overline{EF}$  be two distinct chords of a circle that intersect at midpoint  $M$  of chord  $\overline{AB}$ . If chords  $\overline{CF}$  and  $\overline{ED}$  intersect  $\overline{AB}$  at points  $X$  and  $Y$ , respectively, then  $M$  is also the midpoint of line segment  $\overline{XY}$ .



**Proof:**

To prove the theorem, we begin by constructing perpendiculars  $x_1$  and  $y_1$  from  $X$  and  $Y$  to  $\overline{CD}$ , and perpendiculars  $x_2$  and  $y_2$  from  $X$  and  $Y$  to  $\overline{EF}$ . For convenience, we shall let  $a = AM = MB$ ,  $x = XM$ , and  $y = MY$ . We obtain similar triangles  $Cx_1$  and  $Ey_2$ ,  $Mx_1$  and  $My_1$ ,  $Fx_2$  and  $Dy_2$ ,  $Mx_2$  and  $My_2$ . These similar triangles yield the following:

$$\frac{x}{y} = \frac{x_1}{y_1} = \frac{x_2}{y_2}, \quad \frac{x_1}{y_2} = \frac{CX}{EY}, \quad \frac{x_2}{y_1} = \frac{XF}{YL}$$

$$\frac{x_2^2}{y^2} = \frac{x_1 x_2}{y_1 y_2} = \frac{x_1 x_2}{y_2 y_1} = \frac{(CX)(XF)}{(EY)(YD)} = \frac{(AX)(XB)}{(AY)(YB)} = \frac{(a-x)(a+x)}{(a+y)(a-y)} = \frac{a^2 - x^2}{a^2 - y^2}$$

By setting  $\frac{x_2^2}{y^2} = \frac{a^2 - x^2}{a^2 - y^2}$  and cross multiplying, we get  $a^2 x^2 - x^2 y^2 = a^2 y^2 - x^2 y^2$

which is equal to  $a^2 x^2 = a^2 y^2$ . Dividing each side of the equation by  $a^2$ , yields  $x^2 = y^2$ . Taking the square root on both sides, we see that  $x=y$ . By replacing  $x$  and  $y$  with  $XM$  and  $MY$  respectively, the equation becomes  $XM=MY$ . Therefore  $M$  is the midpoint of  $\overline{XY}$ .



# MATH TEAM

The Stuyvesant High School Math Team, which meets at 8:00 every morning in rooms 517 and 509, has had another successful year. It has won competitions on the city, state and national levels under its coaches Mr. Kurt E. Ritterman and Mr. Richard B. Geller, and captains Andrew Lee and Michael Friedman.

In the New York City Interscholastic Math League competitions, the Senior A Team took first place in the fall term, with Michael Friedman tied for second place and Andrew Lee for third place in the city. The Junior A Team won in both the fall and spring terms, the latter by a considerable margin. In the fall, Junior team captain, Darien Lefkowitz and Tony Friedman tied for first place, and Andrew Chen, Louis Tao, and Elizabeth Wilmer were tied for third place. Spring term, the Senior A Team consisted of captains Andrew Lee and Michael Friedman, David Yuen, Daniel Zuckerman, The B team consists of captain Ezra Peisach, Tao Alter, Chao-Hsiung Cheng, James Liu, and Shuen-Shi Shin. The Junior A Team consists of captain Darien G. Lefkowitz, Antony Friedman, Andy Chen, Lou Tao, and Elizabeth Wilmer; The B Team consists of captain Peter Ying, Chi K. Chan, Khursheed Imam, Daniel Peisach, and Ken Yoon. In addition, Stuyvesant fielded a Junior C team for the first time.

Stuyvesant finished first in the state this year in the New York State Math League Competition, and first in the country in the annual American High School Mathematics Exam, with a total score of 366, out of a possible 450. The team score was comprised of Antony Friedman, 126; Lefkowitz, 123; and Lee, 117. Eleven students scored over 95 and thus qualified for the American Invitational Mathematics Examination. Lefkowitz' score on the AIME qualified him to take the U.S. Mathematics Olympiad, and he has subsequently been invited to the training session at West Point this June for the IMO team.

This year at the annual New York State Math League competition, Stuyvesant was represented by six members of the N.Y.C. Team A, which won the event without event. Several students have been invited to serve on the New York City teams in the American Regions Mathematics League contest to be held at Pennsylvania State University early this June.

Students interested in solving challenging mathematics problems, learning advanced techniques in high school mathematics, and joining Stuyvesant's winning tradition should appear in room 517 during zero period early next term.

# HERON'S FORMULA

Heron was an Alexandrian mathematician. He derived a simple formula for finding the area of a triangle, given the lengths of its sides.

## Derivation I

$$\begin{aligned}
 x + y &= b \\
 \sqrt{a^2 - h^2} + \sqrt{c^2 - h^2} &= b \quad (\text{Pythagorean Theorem}) \\
 \sqrt{a^2 - h^2} &= b - \sqrt{c^2 - h^2} \\
 a^2 - h^2 &= b^2 - 2b\sqrt{c^2 - h^2} + c^2 - h^2 \\
 a^2 - b^2 - c^2 &= -2b\sqrt{c^2 - h^2} \\
 (a^2 - b^2 - c^2)^2 &= 4b^2(c^2 - h^2) \\
 (a^2 - b^2 - c^2)^2 &= (2bc)^2 - (2bh)^2 \\
 (2bh)^2 &= (2bc)^2 - (a^2 - b^2 - c^2)^2 \\
 [4(\frac{1}{2}bh)]^2 &= [2bc - (a^2 - b^2 - c^2)][2bc + (a^2 - b^2 - c^2)] \\
 (4K)^2 &= [b^2 + 2bc + c^2 - a^2][a^2 - (b^2 - 2bc + c^2)] \\
 16K^2 &= [(b+c)^2 - a^2][a^2 - (b-c)^2] \\
 K^2 &= \frac{[(b+c+a)(b+c-a)][(a-b+c)(a+b-c)]}{16} \\
 K &= \sqrt{\left(\frac{b+c+a}{2}\right)\left(\frac{b+c-a}{2}\right)\left(\frac{a-b+c}{2}\right)\left(\frac{a+b-c}{2}\right)} \\
 K &= \sqrt{s(s-a)(s-b)(s-c)}
 \end{aligned}$$

## Derivation II

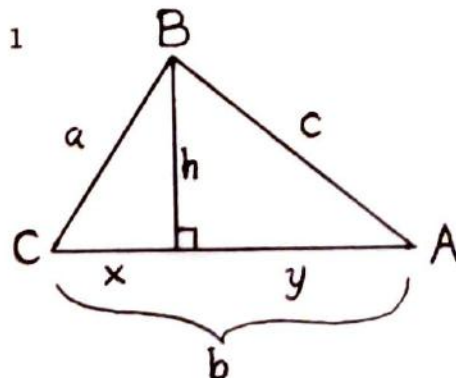
$$\begin{aligned}
 K &= \frac{1}{2}absin C \\
 \frac{2K}{ab} &= sin C \\
 \frac{4K^2}{a^2b^2} &= sin^2 C \\
 cos C &= \frac{a^2+b^2-c^2}{2ab} \quad (\text{Law of Cosines}) \\
 cos C &= \frac{(a+b)^2 - c^2}{2ab} - 1 \\
 cos^2 C &= \frac{[(a+b)^2 - c^2]^2}{4a^2b^2} - \frac{(a+b)^2 - c^2}{ab} + 1
 \end{aligned}$$

$$\begin{aligned}
 sin^2 C + cos^2 C &= 1 \\
 \frac{16K^2}{4a^2b^2} + \frac{[(a+b)^2 - c^2]^2}{4a^2b^2} - \frac{4ab[(a+b)^2 - c^2]}{4a^2b^2} + 1 &= 1
 \end{aligned}$$

$$\begin{aligned}
 16K^2 &= 4ab[(a+b)^2 - c^2] - [(a+b)^2 - c^2]^2 \\
 &= [(a+b)^2 - c^2][4ab - [(a+b)^2 - c^2]] \\
 &= (a+b+c)(a+b-c)[c^2 - (a^2 - 2ab + b^2)] \\
 &= (a+b+c)(a+b-c)[c^2 - (a-b)^2] \\
 &= (a+b+c)(a+b-c)(c-a+b)(c+a-b)
 \end{aligned}$$

$$K = \sqrt{\left(\frac{a+b+c}{2}\right)\left(\frac{b+c-a}{2}\right)\left(\frac{a-b+c}{2}\right)\left(\frac{a+b-c}{2}\right)}$$

$$K = \sqrt{s(s-a)(s-b)(s-c)}$$





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# REGENTS REFRESHER

## Equations of Plane Curves:

Line:  $y = mx + b$  where  $m$  = slope and  $b$  =  $y$  intercept.

$y - y_1 = m(x - x_1)$  where  $(x_1, y_1)$  is a point on the line and  $m$  = slope.

Slopes of parallel lines are equal.

The product of the slopes of two perpendicular lines is  $-1$ .

Circle:  $(x-h)^2 + (y-k)^2 = r^2$ , where  $(h, k)$  is the center of the circle and  $r$  = radius.

Ellipse:  $x^2/a^2 + y^2/b^2 = 1$ , where  $a$  and  $b$  are the major and minor axes.

Parabola:  $(x-h)^2 = 2p(y-k)$ , where  $(h, k)$  is the vertex and  $p$  is the distance from vertex to directrix.  
or  $y = ax^2 + bx + c$ , where  $x = -b/2a$  is the axis of symmetry.

Hyperbola:  $x^2/a^2 - y^2/b^2 = 1$ , where  $a$  and  $b$  are the major and minor axes.

Logarithms:  $\log_B N = E$  is equivalent to  $B^E = N$

$$\log_c 1 = 0$$

$$\log_c c = 1$$

$$\log_c c^b = b$$

$$\log_c a^b = b \log_c a$$

$$\log_c ab = \log_c a + \log_c b$$

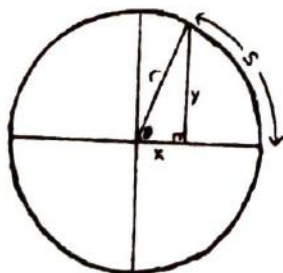
$$\log_c 1/a = -\log_c a$$

$$\log_c (a/b) = \log_c a - \log_c b$$

$$\log_c \sqrt[b]{a} = (\log_c a)/b$$

$$\log_b a = \frac{\log_c a}{\log_c b}$$

## Trigonometry:



$$r = \sqrt{x^2 + y^2}$$

$$s = r\theta$$

SOH-CAH-TOA

$$\sin \theta = y/r$$

$$\csc \theta = r/y = 1/\sin \theta$$

$$\cos \theta = x/r$$

$$\sec \theta = r/x = 1/\cos \theta$$

$$\tan \theta = y/x$$

$$\cot \theta = x/y = 1/\tan \theta$$

$$\sin 0 = \cos(90^\circ - \theta)$$

$$\tan 0 = \cot(90^\circ - \theta)$$

$$\sec \theta = \csc(90^\circ - \theta)$$

$$\sin^2 \theta + \cos^2 \theta = 1 \quad \tan^2 \theta + 1 = \sec^2 \theta$$

$$\cot^2 \theta + 1 = \csc^2 \theta$$



## Trigonometry, cont.

Sum/Difference Formulas:  $\sin(x \pm y) = \sin x \cos y \pm \sin y \cos x$   
 $\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$   
 $\tan(x \pm y) = \frac{\tan x \pm \tan y}{1 \mp \tan x \tan y}$

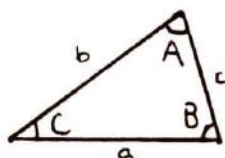
Double Angle Formulas:  $\sin(2x) = 2\sin x \cos x$   
 $\cos(2x) = \cos^2 x - \sin^2 x = 2\cos^2 x - 1 = 1 - 2\sin^2 x$   
 $\tan(2x) = \frac{2\tan x}{1 - \tan^2 x}$   
 (Note: the double angle formulas can be derived from the sum formulas by letting  $x=y$ .)

Half Angle Formulas:  $\sin(x/2) = \pm \sqrt{\frac{1 - \cos x}{2}}$        $\cos(x/2) = \pm \sqrt{\frac{1 + \cos x}{2}}$   
 $\tan(x/2) = \pm \sqrt{\frac{1 - \cos x}{1 + \cos x}}$

In the graph of  $y = n \sin rx$ , the amplitude =  $n$   
 the period =  $360^\circ / r$

$360^\circ = 2\pi$  radians      To convert degrees to radians, multiply by  $\pi/180^\circ$   
 To convert radians to degrees, multiply by  $180^\circ/\pi$

## Triangles:



Area =  $(1/2)ab \sin C$

Law of Sines:  $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = \text{diameter of circumcircle}$

Law of Cosines:  $c^2 = a^2 + b^2 - 2ab \cos C$

Law of Tangents:  $\frac{\tan((A+B)/2)}{\tan((A-B)/2)} = \frac{a+b}{a-b}$

## Laws of Exponents:

$x^m \cdot x^n = x^{m+n}$

$(x^m)^n = x^{mn}$

$x^m / x^n = x^{m-n}$

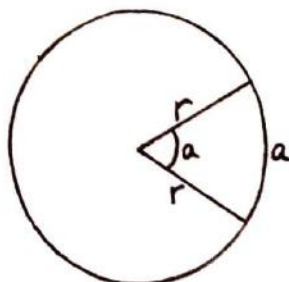
$x^{-m} = 1/x^m$

$x^0 = 1$

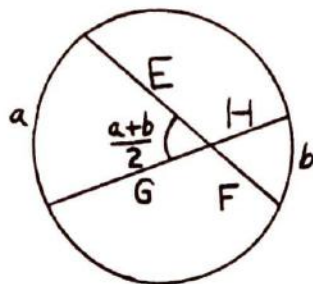
$x^{1/m} = \sqrt[m]{x}$

$x^{m/n} = \sqrt[n]{x^m} = (\sqrt[n]{x})^m$

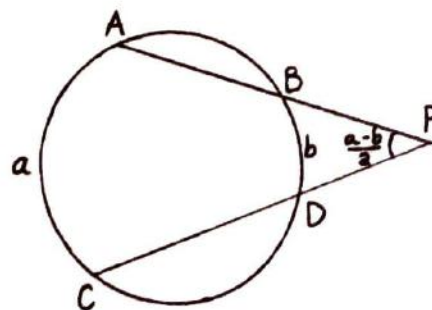
## Geometry:



$r = \text{radius}$

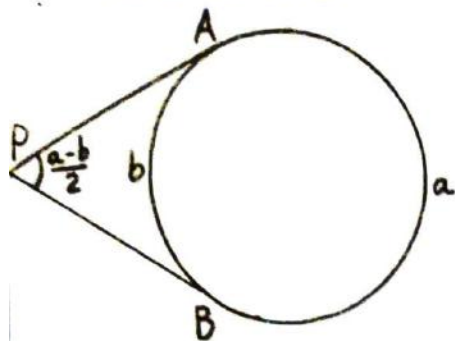


$EF = GH$

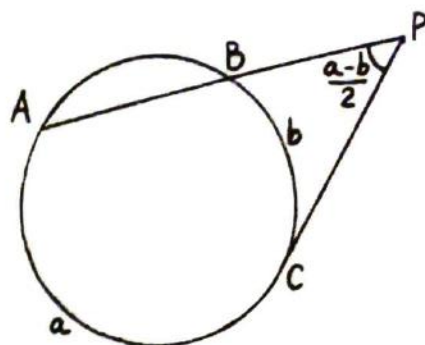
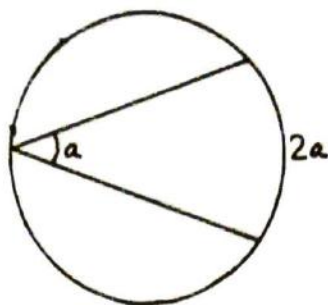


$(PA)(PB) = (PC)(PD)$

Geometry, cont.



$$PA=PB$$



$$(PA)(PB)=(PC)^2$$

The required proofs for the Geometry Regents may be found in the Dressler text, pp. 733-737. The following are the numbers of the required proofs; 3,5,6,7,8,9,10,11,12,13.

(A=Angle, S=Side, b=base, h=altitude)

Two triangle are congruent if you can prove SSS, SAA, SAS, or A&A.

Two triangles are similar if you can prove AA.

Two right triangles are congruent if you can prove their hypotenuses are congruent and one pair of legs are congruent.

The sum of the angles in a triangle is  $180^\circ$ .

The base angles of an isosceles triangle are congruent.

Area of triangle=  $(1/2)bh$ .

Area of a trapezoid=  $(1/2)(b_1+b_2)h$ .

Area of a parallelogram=  $bh$ .

Area of a regular n-gon=  $(1/2)ap$  (where a=apothem, p=perimeter).

In a right triangle with legs a and b and hypotenuse c,  $a^2 + b^2 = c^2$ .

In a circle, the diameter that is perpendicular to a chord bisects the chord.

The locus of points at distance r from point P is the circle with center P and radius r.

The locus of points equidistant from two points P and Q is the perpendicular bisector of  $\overline{PQ}$ .

The locus of points equidistant from two lines l and m are the bisectors of the angles formed by l and m.

(B=area of base, h=haight, r=radius)

Volume of a prism=  $Bh$

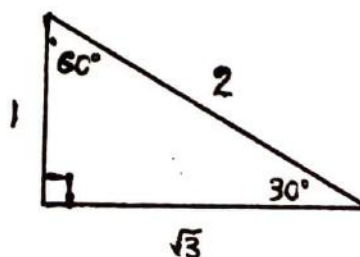
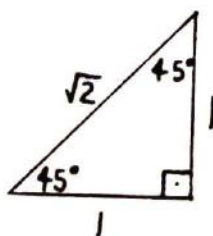
Volume of a cylinder=  $\pi r^2 h$

Volume of a pyramid=  $(1/3)Bh$

Volume of a cone=  $(1/3)\pi r^2 h$

Volume of a sphere=  $(4/3)\pi r^3$

The required constructions for the Geometry Regents may be found in the Dressler text, pp. 616-639. The following are the numbers of the required constructions; 5,6,7,12,13,18,19,20,21,22,25,27,28,29





# FIBONACCI

# RATIOS

The famous Fibonacci Sequence, which begins 1,1,2,3,5,8,13,...etc , is given by the recursive definition:

$$f_n = f_{n-1} + f_{n-2}, \quad f_1 = f_2 = 1$$

This sequence was originally developed by Fibonacci<sup>1</sup> to solve a rabbit generation problem.<sup>2</sup> However it is also of great interest to mathematicians as it corresponds to many other natural phenomena.

Another related sequence of interest is the Lucas Sequence, 1,3,4,7,11,18,29,47,76,... etc. It can be defined recursively:

$$L_n = L_{n-1} + L_{n-2}, \quad L_1 = 1 \quad L_2 = 3$$

Interestingly, which can be proved by induction.

A formula which I personally have discovered is concerned with the quotient of any two terms of the Fibonacci Sequence. I happened upon it one day by dividing a series of numbers, and finding that there appeared to be a definite pattern.

The formula:

$$\frac{f_{x+n}}{f_x} = L_n + (-1)^{n+1} \cdot \frac{f_{x-n}}{f_x}$$

where  $f_x$  and  $f_{x+n}$  are any two terms and  $n$  is the number of terms between them.

When I first discovered this formula, I verified it on my computer, and then attempted to prove the general case. The method I used was a variant of the method of mathematical induction.

The usual method of induction involves three steps: One, proving the case when  $n=1$ ; Two, assuming the case for  $n=k$ ; and Three, using the assumption in step Two proving the case for  $n=k+1$ . This proves that  $n$  belongs to the set of natural numbers.

Since the usual method did not work, I tried a variation of mathematical induction, which involved:

- (1) Proving the cases for  $n=1$  and  $n=2$ ,
- (2) Assuming for the cases  $n=k$ , and  $n=k+1$
- (3) Proving the case for  $n=k+2$

If I can prove step 3 using the assumption in step 2, I essentially have proved that  $n$  belongs to the set of natural numbers. This is intuitively correct, since if 1 and 2 works, 3 works; if 2 and 3 works, 4 works and so on.

Here is the proof:

1.  $n = 1$

$$\begin{aligned}\frac{f_{x+1}}{f_x} &= L_1 + (-1)^{1+1} \frac{f_{x-1}}{f_x} \\ \frac{f_{x+1}}{f_x} &= 1 + \frac{f_{x-1}}{f_x} \\ \frac{f_{x+1}}{f_x} &= \frac{f_x}{f_x} + \frac{f_{x-1}}{f_x} \quad \left( \frac{f_x}{f_x} = 1, \text{ Provided } f_x \neq 0 \right)\end{aligned}$$

multiplying both sides by  $f_x$ , we get  $f_{x+1} = f_x + f_{x-1}$   
which is the basic definition of the sequence.

$n = 2$ .

$$\begin{aligned}\frac{f_{x+2}}{f_x} &= L_2 + (-1)^{2+1} \frac{f_{x-2}}{f_x} \\ \frac{f_{x+2}}{f_x} &= 3 - \frac{f_{x-2}}{f_x} \\ \frac{f_{x+2}}{f_x} &= \frac{3f_x}{f_x} - \frac{f_{x-2}}{f_x} \quad \left( 3 \frac{f_x}{f_x} = 3 \cdot 1 = 3 \right)\end{aligned}$$

multiplying by  $f_x$ , we get  $f_{x+2} = 3f_x - f_{x-2}$

but  $f_x = f_{x-1} + f_{x-2}$  (from definition)

so

$$\begin{aligned}f_{x+2} &= 2f_x + f_{x-1} + f_{x-2} - f_{x-2} \\ &= 2f_x + f_{x-1} \\ &= f_x + f_{x+1} \quad (\text{from definition}) \\ f_{x+2} &= f_{x+2} \quad (" \quad " \quad ")\end{aligned}$$

2. Assume that

$$n=k \quad \frac{f_{x+k+2}}{f_x} = L_k + (-1)^{k+1} \frac{f_{x-k}}{f_x}, \text{ AND}$$

$$n=k+1 \quad \frac{f_{x+k+3}}{f_x} = L_{k+1} + (-1)^{k+2} \frac{f_{x-k-1}}{f_x} \text{ ARE TRUE}$$



3. We have to prove that the case of  $n = k + 2$  has to be true.

$$\begin{aligned}
 \frac{f_{x+k+2}}{f_x} &= \frac{f_{x+k}}{f_x} + \frac{f_{x+k+1}}{f_x} \\
 &= L_k + (-1)^{k+1} \frac{f_{x-k}}{f_x} + L_{k+1} + (-1)^{k+2} \frac{f_{x-k-1}}{f_x} \\
 &= L_k + L_{k+1} + (-1)^{(k+1)} \frac{f_{x-k}}{f_x} + (-1)^{k+2} \frac{f_{x-k-1}}{f_x} \\
 &= L_{k+2} + (-1)^{k+2} \left( -\frac{f_{x-k}}{f_x} + \frac{f_{x-k-1}}{f_x} \right) \\
 &= L_{k+2} + (-1)^{k+2} (-1) \left( \frac{f_{x-k} - f_{x-k-1}}{f_x} \right) \\
 &= L_{k+2} + (-1)^{k+3} \frac{f_{x-k-2}}{f_x} \quad (\text{FROM DEFINITION}) \\
 \frac{f_{x-k}}{f_x} &= \frac{f_{x-k-1}}{f_x} + \frac{f_{x-k-2}}{f_x}
 \end{aligned}$$

Q.E.D.

#### USEFUL TERMS, continued

14. "The proof is beyond the scope of this book ..."  
(We don't know how to prove it.)
15. "The proof is left to the student ..."  
(Welcome to mental S&M.)
16. " $\exists \alpha \exists \phi < \theta$  ( $\theta < \pi$ );  $|\beta - \alpha| < \delta$  ..."  
(It's time to consider majoring in Greek.)
17. "Q.E.D."  
(=T.G.W.A.D., or sometimes T.G.I.F. In any case,  
Q.E.D. ≠ quite easily derived.)

-with thanks to David Zagorski  
and 18.03 at M.I.T.

# FUNDAMENTAL THEOREM OF ALGEBRA

The Fundamental Theorem of Algebra was suspected for several centuries before K.F. Gauss proved it, perhaps half a dozen times, beginning in 1799. It had been shown that the field of real numbers ( $R$ ) is metrically complete; i.e. every real sequence convergent by Cauchy's criterion converges on an element of  $R$ . The Fundamental Theorem states that the field of complex numbers ( $C: \{a+bi: a, b \in R\}$ ) is algebraically complete; this is actually a statement about the ring  $C[z]$  of polynomials over  $C$ . Before undertaking the proof we require some preliminary results.

If  $z = x+yi$ , we define:  $\bar{z}$ , the conjugate of  $z$ , by  $\bar{z} = x-yi$   
 $|z|$ , the absolute value of  $z$ , by  $|z| = \sqrt{z\bar{z}} = \sqrt{x^2+y^2}$

We also need two theorems on absolute values whose proofs we leave to the reader:

For all  $a, b \in C$ ,  $|a||b| = |ab|$

ii)  $|a| + |b| \geq |a+b|$ , with equality iff  $a/b$  belongs to  $R$  or  $b=0$

Note, from (ii), that  $|a+b| + |-b| \geq |a|$ ; yielding  $|a+b| \geq |a| - |-b| = |a| - |b|$ .  
 Thus, (iii)  $|a| + |b| \geq |a+b| \geq |a| - |b|$  (the triangle inequality)

Lemma 1: If  $p(z) \in C[z]$ , then  $p(z)$  and  $|p(z)|$  are continuous over  $C$ .

Continuity and the limit are defined over  $C$  in exactly the same manner as over  $R$ .

The reader may wish to prove the lemma by observing:

a.) The identity function  $I: z \rightarrow z$  and the absolute value function  $T: z \rightarrow |z|$  are continuous, and

b.) if  $f$  and  $g$  are continuous, then  $f+g$ ,  $fg$ , and  $f \circ g$  are continuous.

Lemma 2: Given any polynomial  $P(z) = \sum_{n=0}^N a_n z^n$ , there exists an  $M$  in  $R^+$  such that  $|z| > M$  implies  $|p(z)| > |a_n z^n|/2$  and  $|p(z)| > |p(0)|$ .

proof: Let  $a = \max_i |a_i/a_n|$ , and  $M = \max \{1, 2a(n-1), |2p(0)/a_n|\}$ ; let  $q(z) = p(z) - a_n z^n$ .  
 If  $|z| > M$ , we first show  $|a_n z^n|/2 > |q(z)|$ .

Since  $|z| > 1$ ,  $|z^{n-j}| \geq |z| > 2a(n-1)$ , for  $j=1, 2, \dots, n-1$ .

thus  $|z^n| > 2a(n-1)|z^j| \geq 2(n-1)|a_j/a_n||z^j|$

and  $|a_n z^n|/2(n-1) > |a_j z^j|$ .



Adding the inequalities for  $j=1,2,\dots,n-1$ ,

$$|a_n z^n / 2| > \sum_{j=1}^{n-1} a_j z^j$$

By the triangle inequality, iterated  $(n-2)$  times,  $|a_n z^n / 2| > |q(z)|$

$$\text{or } |a_n z^n| - |q(z)| > |a_n z^n / 2|$$

Now,  $|p(z)| \geq |a_n z^n| - |q(z)|$ , also by the triangle.  $\neq$  Therefore  $|p(z)| > |a_n z^n / 2|$

Note also  $|p(z)| > |a_n z^n / 2| > |a_n z / 2| > |a_n M / 2| \geq |a_n (2p(0)/a_n) / 2| = |p(0)|$

Q.E.D. (lemma 2)

Theorem (Gauss): Every non-constant polynomial with coefficients in  $C$  has a zero in  $C$ .

(Fun. Th'm. of

Algebra)

corollary: Any polynomial  $p(z) \in C[z]$  of degree  $n \geq 1$  may be factored into  $n$  linear factors.

The corollary follows directly from the theorem by the use of the remainder theorem and downward induction.

proof of theorem: It has been shown that there exists some circle  $P$ , centred about the origin on the Gaussian plane, of some finite radius  $M$ , such that, for every point  $z$  outside of  $P$ ,  $|p(z)|$  is greater than  $|p(0)|$ . Since the closed disk possesses the topological property of compactness, and is "simply connected", the continuous function  $|p(z)|$  attains an absolute minimum on the disk. Say the minimum occurs at  $z_0$ . Then  $|p(z_0)|$  is an absolute minimum over  $C$ . Consider  $q(z) = p(z + z_0)$ . This function is the same as  $p(z)$  with  $z_0$  translated to the origin;  $|q(z)|$  is minimized at 0, and it remains to show  $q(0) = 0$ .

Assume  $q(0) = \alpha \neq 0$ ; we will now produce a point  $\omega$ , for which  $|q(0)| > |q(\omega)|$ , contradicting the fact that  $|\alpha|$  is minimal.

We know  $q(z) = \alpha + \beta z^m + c_{m+1} z^{m+1} + \dots + c_n z^n$

where  $\beta$  is the first non-zero coefficient when the terms are written in ascending order. Let  $r$  be any value of  $(-\alpha/\beta)^{1/m}$ .

$$\begin{aligned} q(rz) &= \alpha - \alpha z^m + z^m (d_1 z + d_2 z^2 + \dots + d_{n-m} z^{n-m}) \\ &= \alpha \left[ 1 - z^m + z^m \left( \frac{d_1 z}{\alpha} + \dots + \frac{d_{n-m} z^{n-m}}{\alpha} \right) \right] \end{aligned}$$

for some complex  $d_1, d_2, \dots, d_{n-m} \in C$

Let  $r(z) = (d_1 z + d_2 z^2 + \dots + d_{n-m} z^{n-m}) / \alpha$ ; since  $r(z) \in C[z]$  is continuous,  $\lim_{z \rightarrow 0} r(z) = 0$ .

Thus there exists a  $\delta$  in  $R^+$  such that  $|z| < \delta$  implies  $|r(z)| < 1$ . Choose  $z$  as a positive real with  $z < \delta$  and  $z < 1$ .

$$\begin{aligned} \text{then } |q(rz)| &= |\alpha| |1 - z^m + z^m r(z)| \leq |\alpha| |1 - z^m| + |\alpha| |z^m| \\ &< |\alpha| |1 - z^m| + |\alpha| |z^m| = |\alpha| (1 - z^m) + |\alpha| (z^m) \\ &= |\alpha| \end{aligned}$$

Therefore  $|q(rz)| < |q(0)|$ , contradicting the fact that  $|q(0)| \geq |q(rz)|$ .

Thus the assumption was false, and  $\alpha = q(0) = p(z_0) = 0$ .

Q.E.D.

# IRRATIONALITY OF $\pi$

It is well known that the number  $\pi$  is irrational- that is, it cannot be represented as a ratio  $p/q$ , where  $p$  and  $q$  are integers. However, the proof of this fact is often not easy. In this article, we shall examine one proof of  $\pi$ 's irrationality that uses the following lemmas, the validity of which may be demonstrated through repeated integration by parts.

Lemma 1: If  $g, g', g^{(iv)}, \dots, g^{(n-2)}$  all vanish at 0 and  $\pi$ , and  $n$  is even, then  $\int_0^\pi g(x) \sin x \, dx = (-1)^{n/2} \int_0^\pi g^{(n)}(x) \sin x \, dx$ .

Lemma 2: If  $g$  is a polynomial and all of its derivatives have integral values at 0 and  $\pi$ , then  $\int_0^\pi g(x) \sin x \, dx$  is also an integer.

We will use the indirect method of proof here; we will assume  $\pi$  to be rational and prove a contradiction. Therefore, assume  $\pi = p/q$ ,  $p, q \in \mathbb{Z}$ .

First, consider  $f(x) = 1/n! [x(qx-p)]^n$ , where  $n$  is even. Note that  $f$  vanishes at 0 and  $\pi$ , and that it is positive for all  $x$  on the interval  $(0, \pi)$ . Clearly,  $0 < \int_0^\pi f(x) \sin(x) \, dx < \int_0^\pi f(x) \, dx$ . Also, by the arithmetic-geometric mean inequality,  $f(x) \leq 1/n! (p^2/4q)^n$ . Thus, we have that

$0 < \int_0^\pi f(x) \sin(x) \, dx \leq \int_0^\pi f(x) \, dx \leq \int_0^\pi 1/n! (p^2/4q)^n \, dx$ . The reader should note that  $\lim_{n \rightarrow \infty} 1/n! (p^2/4q)^n = 0$ .

Now, we rewrite  $f(x)$  as

$$\begin{aligned} f(x) &= x^n / n! (qx-p)^n \\ &= x^n / n! \sum_{k=0}^n \binom{n}{k} (qx)^k (-p)^{n-k} \\ &= \sum_{k=0}^n 1/n! \binom{n}{k} x^{n+k} q^k (-p)^{n-k} \end{aligned}$$

It is left to the reader to verify that  $f$  satisfies Lemmas 1 and 2 (enjoy it!). Our contradiction arises immediately from the fact that  $\int_0^\pi f(x) \sin x \, dx$  takes on integral values for all  $n$ , yet is between 0 and 1 for  $n$  sufficiently large. Thus, the assumption that  $\pi = p/q$ ,  $p, q \in \mathbb{Z}$  was incorrect;  $\pi$  is irrational.

Q. E. D.



Two gods ( call them A and B ) once played a series a games of poker. They sat down at noon, each with a stack of dollar bills containing one dollar for each natural number. The serial number of A's bills were 1, 3, 5, 7 ..., while B's were numbered 2, 4, 6, 8... They agreed that before each hand, B would pay A one dollar, and after each hand, A would give B two dollars if he lost, and nothing if he won.

The first hand was played at noon, the second at 12:30, the third at 12:45, and so on. When the session was over at 1:00pm., B remarked smugly that he had won every game. A agreed, but it was soon discovered that A, who had lost every game, now possessed all the money ! For in hand 1, B had given him bill #1 and gotten back #2 and #4. In the second hand, #2 was returned to A, and he gave B #6 and #8; in general, A received #n in round n, and gave B #(4n-2) and #4n. In fact, every bill which A was eventually returned and kept.

The next day, the two gods met again at noon, to play another game. They took turns adding terms to a decreasing sequence of positive numbers. A went first at noon, and they alternated at 12:30, 12:45, and so on, as before. They agreed that A was to win if the sum of the numbers converged to a rational or diverged, and B was to win if the sum converged to an irrational. Assuming best play for both sides, who should have been declared the winner at one o'clock?

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# KING CHICKEN THEOREMS

The king chicken theorems deal with a mathematical model for a non-transitive pecking order on a flock of chickens. The sociological applications are dubious, but the model itself exhibits interesting properties.

By a flock, we mean a non-empty finite set  $F$  of elements called Chickens, together with a pecking relation in which, for every pair of chickens, exactly one pecks the other. A chicken  $k$  is a king if, for all  $c \in F$ , either  $k$  pecks  $c$  or there exists a chicken  $b$  for which  $k$  pecks  $b$  and  $b$  pecks  $c$ . (Chickens have a relatively short memory.) In addition, a chicken is a tyrant if he pecks every other chicken in the flock.

Formally, we represent the pecking order by a multiple-valued function  $P$  which maps each  $c \in F$  onto a subset of  $F_c = F - \{c\}$ :

$$Pc = \{x \in F_c : c \text{ pecks } x\}.$$

If  $A \subseteq F$ , then the image of  $A$  under  $P$  is  $PA = \bigcup_{a \in A} Pa$ . The first condition given above can be restated as

$$Pa \cup P^{-1}a = F_a, \text{ and } Pa \cap P^{-1}a = \emptyset, \text{ for all } a \in F, \text{ or simply}$$

$$F_a - Pa = P^{-1}a.$$

The definition of a king is written concisely as

$$P^{-1}k \subseteq P^2k = P(Pk),$$

and the definition of a tyrant becomes  $Pt = F_t$ .

A simple graphical representation has each chicken a vertex of a partially directed graph, with "a pecks b" symbolized as " $a \rightarrow b$ ".

Here is a flock with only one king (a tyrant): 

Here are three flocks in which every chicken is a king:

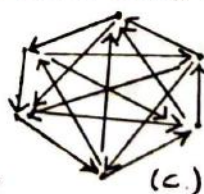


Figure 1.

**Theorem 1:** Every Flock has a king.

proof: It is clear that flocks of size 1 and 2 must have kings. We now induct on the size of the flocks. If a flock contains  $n$  chickens including a king  $k$ , but is a subflock of an  $(n+1)$ -flock in which  $k$  is not a king, then the  $(n+1)$ -st chicken  $k'$  is a king. This follows from the observation that  $k'$



pecks  $k$  and every chicken pecked by  $k$ , and that each of the remaining chickens is pecked by some chicken pecked by  $k$ . This completes the induction.

Theorem 2: If a chicken is pecked, then he is pecked by a king.

Proof: For  $c \in F$ , we wish to show that if  $P^{-1}c$  is non-empty, then it contains a king. By theorem 1,  $P^{-1}c$  contains a local king under the restriction of  $P$  to the subflock  $P^{-1}c$ . It is obvious that this local king as well is a global king.

Note that, in general, a local king is not necessarily a global king; nor is a global king necessarily a local king in a subflock to which he belongs.

Theorem 3: If  $c \in F$  is not a king, then some other chicken in  $F$  pecks more chickens than  $c$  does.

Proof: Since  $c$  is not a king, there exists a chicken  $d$  such that  $d$  pecks  $c$  and  $d$  pecks every chicken pecked by  $c$ .

Corollary: If a chicken pecks at least as many chickens as any other member of the flock, then he is a king.

The converse is false; in figure 2,  $k_1$  pecks only  $k_2$ , while  $k_2$  pecks most of the flock.



Theorem 4: If a flock has only one king, then he is a tyrant.

Proof: Since the king is not pecked by another king, he is not pecked at all (see theorem 2). Thus he pecks everyone else in the flock. Conversely, of course, a flock with a tyrant contains no other king.

Extending this idea, one sees that if a flock does contain several kings, then each king must be pecked, and must then be pecked by another king. In particular, we have

Theorem 5: No flock contains precisely 2 kings.

The following theorem is our first non-trivial result.

Theorem 6: There exist flocks of all finite orders, except 2 and 4, in which every chicken is a king.

Proof: We claim that an  $N$ -flock possessing the desired property can be used to generate an  $(N+2)$ -flock which also has every member a king. The theorem will follow from induction, beginning with the diagrams in figure 1.

Now, take an  $N$ -flock  $F$  with a pecking order  $P$  such that every chicken in  $F$  is a king. Construct the  $(N+2)$ -flock  $F^* = F \cup \{a, b\}$  with the pecking order  $P^*$  defined by

$$\left. \begin{aligned} P^*_a &= \{b, c\} \\ P^*_b &= P_c \cup P^{-1}_c = F_c \\ P^*_c &= P_c \cup \{b\} \\ P^*_g &= P_g \cup \{a\}, \text{ for } g \text{ other than } a, b, c. \end{aligned} \right\} \quad \begin{array}{l} \text{where } c \text{ is any particular} \\ \text{chicken in } F. \end{array}$$

It is easy to verify that  $a$  and  $b$  are kings, and that the other chickens remain kings, while the pecking order remains the same among the members of the original flock. Note how the order  $P^*$  (figure 3) was obtained from figure 1b above.

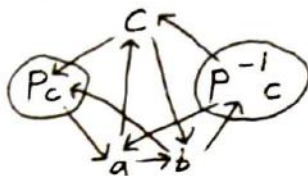


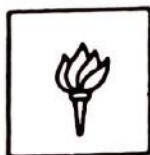
figure 3

As stated before, induction gives us  $N$ -flocks composed entirely of kings for all odd  $N$ , and for even  $N \geq 6$  (see figure 1c).

Let's define a chicken  $c$  to be a serf if for every other chicken  $d \in F$ , either  $d$  pecks  $c$  or there is a chicken  $b$  for which  $d$  pecks  $b$  and  $b$  pecks  $c$ . Also, a eunuch is a chicken pecked by every member of the flock. Every true theorem, including those above, has a dual in which "king", "pecks", "serf", "tyrant", "is pecked by", "eunuch", etc., are changed to, respectively, "serf", "is pecked by", "king", "eunuch", "pecks", "tyrant", etc. (Why?)

See if you can determine necessary and sufficient conditions for a king to be also a serf. For what values of  $n$  do there exist  $n$ -flocks composed entirely of serf-kings?

For more information on this fascinating topic, consult Dr. Stephen B. Maurer's paper in Mathematics Magazine, March, 1980.



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1. Simplify

$$\frac{(a+b+c)^5 - a^5 - b^5 - c^5}{(a+b+c)^3 - a^3 - b^3 - c^3}$$

2. A graph is said to be integrable if one can color its nodes black and white in such a way that at least half the neighbors of any node have the opposite color.

Show that every graph is integrable.

3. A rearrangement (permutation) can change the sum of an infinite series. E.g. if

$$S = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \dots$$

$$S' = 1 + \frac{1}{3} - \frac{1}{2} + \frac{1}{5} + \frac{1}{7} - \frac{1}{4} + \frac{1}{9} + \frac{1}{11} - \frac{1}{6} \dots$$

then

$$S' = S + \frac{2}{1 \cdot 2 \cdot 3} + \frac{2}{5 \cdot 6 \cdot 7} + \frac{2}{9 \cdot 10 \cdot 11} + \dots$$

Show that if a permutation changes the sum of some infinite series then there is a convergent series which that same permutation will change into a divergent series.

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